Microsoft

GitHub Copilot Fundamentals [GH-300]



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About this course

Prerequisites

- Familiarity with integrated development environments (IDEs) and version control systems like GitHub.
- Foundational knowledge of AI and machine learning principles.

Audience profile

Audience profile for this course is the following:

- Developers looking to enhance their coding efficiency with AI-powered tools.
- Programmers interested in learning about responsible AI usage and ethical standards.
- Software engineers seeking to integrate GitHub Copilot into their development workflows.

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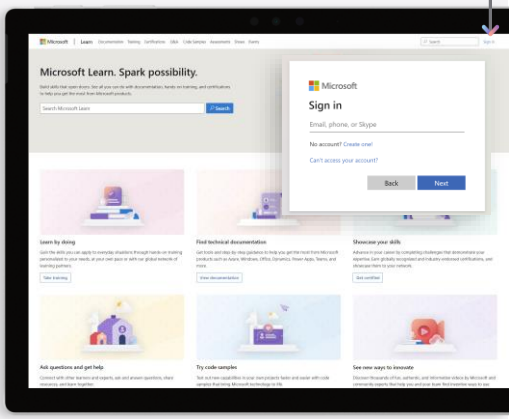
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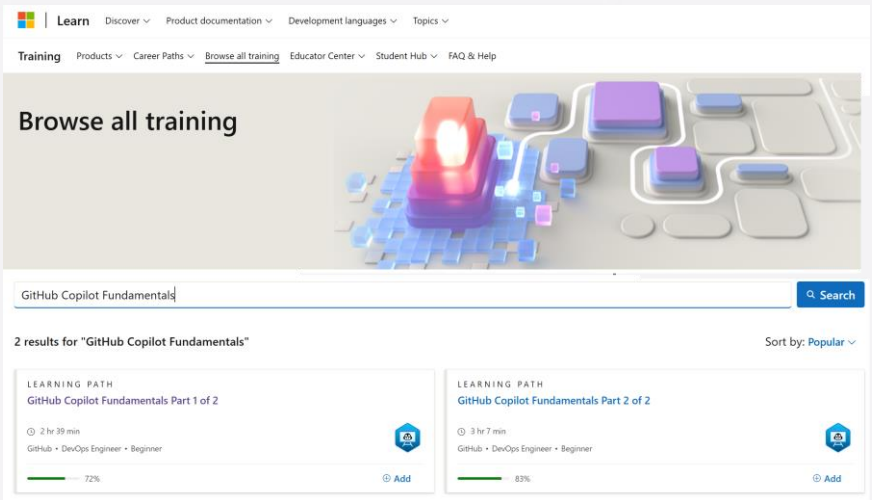


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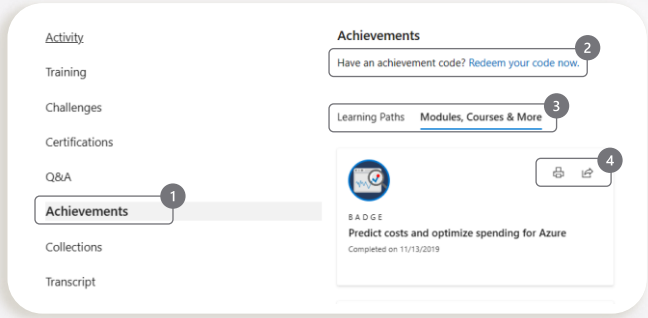
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Get ready for your Microsoft Certification exam

GitHub Copilot

Understand the skills measured by the exam

- Responsible AI
- GitHub Copilot plans and features
- How GitHub Copilot works and handles data
- Prompt Crafting and Prompt Engineering
- Developer use cases for AI
- Testing with GitHub Copilot
- Privacy fundamentals and content exclusions

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Module 1: Responsible AI with GitHub Copilot

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Mitigate AI risks

Opportunities and Risks of AI

- **Innovation and Efficiency**: AI offers significant potential for driving innovation and improving efficiency across various sectors.
- **Risks**: AI systems can lead to unintended consequences such as privacy violations and lack of transparency.

Mitigating AI Risks

- **Governance Frameworks**: Implementing robust governance frameworks is essential to manage AI risks effectively.
- **Transparency and Oversight**: Ensuring transparency in AI processes and incorporating human oversight can help mitigate potential negative impacts.

Responsible AI Principles

- **Ethical Development**: Responsible AI focuses on developing AI systems in a safe, trustworthy, and ethical manner.
- **Human-Centered Design**: Keeping human goals at the center of AI system design decisions ensures fairness, reliability, and transparency.

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Microsoft and GitHub's five principles of responsible AI

- **Fairness**: AI systems should treat all people fairly, avoiding differential impacts on similarly situated groups.
- **Reliability and Safety**: AI systems must operate reliably, safely, and consistently, minimizing unintended harm and perform as intended.
- **Privacy and Security:** Protecting user privacy and data security is crucial, involving consent, minimal data collection, anonymization, and strong encryption.
- **Inclusiveness**: AI systems should be accessible and empower everyone, ensuring they work well for diverse users and are available worldwide.
- **Transparency and Accountability**: AI systems must be understandable, with clear explanations, auditability, and continuous monitoring to ensure responsible operation.

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Module 2: Introduction to GitHub Copilot



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GitHub Copilot, your AI pair programmer

Copilot for Chat

- **Chat Interface**: Integrates with VS Code and Visual Studio.
- **Error Recognition**: Identifies code errors and suggests fixes, generating unit tests.

Copilot for Pull Requests

- **Iterate and Validate**: Integrate suggested changes to code by using Copilot Workspace.
- **Review and Modify**: Developers can review and adjust the suggested descriptions.

Copilot for the CLI

- **Command Composition**: Helps compose commands and loops, making terminal usage more efficient.
- **Syntax Assistance**: Provides obscure find flags and command syntax reminders.

Subscription Plans

- **Personal Accounts**: Available through GitHub Copilot Free and Pro.
- **Organization and Enterprise Accounts**: Available through GitHub Copilot Business and Enterprise.

GitHub Copilot Pro+

- Highest level of access for individual developers.
- Includes Copilot coding agent

GitHub Copilot Business

- **Access Control**: Allows organizations to manage who can use GitHub Copilot.
- **Productivity and Security**: Features include code completions, chat in IDE and mobile, and security filters.

GitHub Copilot Enterprise

- **Enhanced Features**: Includes all Business features plus personalized suggestions based on internal code.
- **Integration and Documentation**: Provides chat interface integration and helps build documentation.

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Interact with Copilot

- **Real-Time Code Completions:** Copilot offers inline suggestions as you type, predicting your next steps and displaying them subtly.
- **Accepting and Rejecting Suggestions:** Use the Tab or right arrow key to accept suggestions, and the Esc key to reject them. Keyboard shortcuts:
  macOS: Command+→
  Windows or Linux: Control+→
- **Command Palette Access:** Quickly access Copilot functions with **Ctrl+Shift+P** (Windows/Linux) or **Cmd+Shift+P** (Mac) to perform tasks like generating unit tests.
- **Interactive Copilot Chat:** Communicate with Copilot using natural language to ask questions or request code snippets directly in your IDE.
- **Context-Specific Inline Chat:** Request code modifications or explanations without switching contexts using **Ctrl+I** (Windows/Linux) or **Cmd+I** (Mac).
- **Comments to Code:** Convert comments into code using natural language processing, making it easy to draft straightforward tasks quickly.
- **Multiple Code Suggestions:** Explore different coding approaches by cycling through multiple alternatives using the light bulb icon or **Alt+/Option+]**.
- **Automated Test Generation:** Generate unit tests for functions or classes to ensure code quality and reliability, saving time and effort.

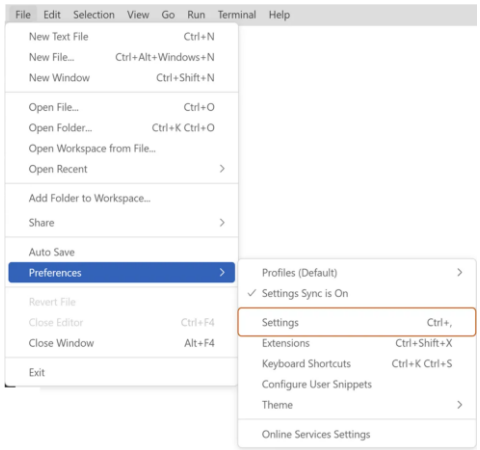
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Set up, configure, and troubleshoot GitHub Copilot

Before you can start using GitHub Copilot, you need to set up a free trial or subscription for your account.

- **Installation:** Install the GitHub Copilot extension from the Visual Studio Marketplace, open VS Code, and sign in to GitHub if prompted.
- **Enable/Disable:** Use the status icon in the bottom pane of VS Code to enable or disable GitHub Copilot globally or for specific languages.
- **Inline Suggestions:** Configure inline suggestions by navigating to Preferences > Settings > Extensions > GitHub Copilot, and selecting or clearing the checkbox for auto completions.



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Module 3: Introduction to prompt engineering with GitHub Copilot

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Prompt engineering foundations and best practices

What is Prompt Engineering?

- Craft clear instructions to guide AI systems like GitHub Copilot.
- Ensures code is syntactically, functionally, and contextually correct.
- Generates context-appropriate code tailored to project needs.

Principles of Prompt Engineering

- Focus on a single, well-defined task or question for clarity.
- Provide explicit and detailed instructions for precise code suggestions.
- Keep prompts concise to maintain clarity without overloading the AI.

Best Practices in Prompt Engineering

- Use descriptive filenames and keep related files open for rich context.
- **Iterate**: Refine prompts based on feedback to improve AI responses.
- **Contextualize**: Provide relevant background information to enhance code suggestions.

Advanced Strategies

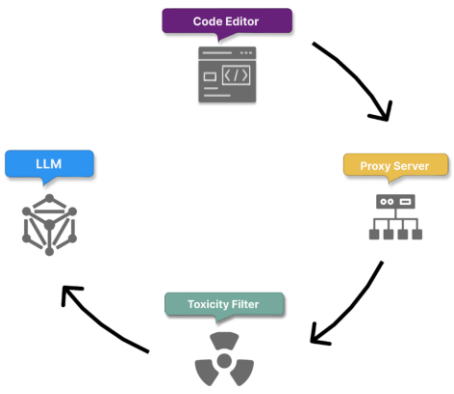
- **Layered Prompts**: Break complex tasks into smaller, manageable prompts.
- **Feedback Loop**: Continuously evaluate and adjust prompts for optimal performance.
- **Collaborative Input**: Involve team members to ensure comprehensive and effective prompts.

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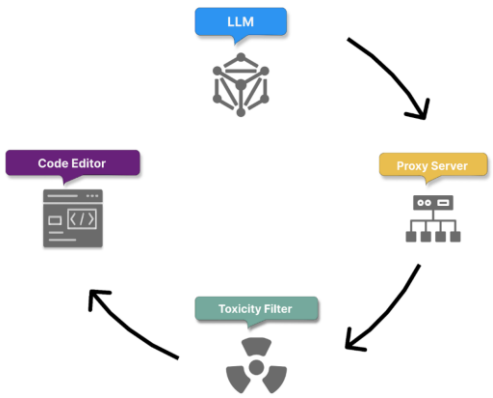
GitHub Copilot user prompt process flow

GitHub Copilot receives prompts and returns code suggestions or responses in its data flow. This process suggests an inbound and outbound flow.

Inbound Flow



Outbound Flow



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GitHub Copilot data

Data Handling for Code Suggestions

- **No Retention**: GitHub Copilot in the code editor discards prompts once a suggestion is returned.
- **Opt-Out Option**: Individual subscribers can opt-out of sharing their prompts for model fine-tuning.

Data Handling for Copilot Chat

- **Response Formatting**: Formats responses for optimal presentation, highlighting code snippets.
- **User Engagement**: Allows follow-up questions and maintains conversation history for context.
- **Data Retention**: Retains prompts and suggestions for 28 days outside the code editor.

Prompt Types Supported by Copilot Chat

- **Direct Questions**: Ask specific coding questions or troubleshoot issues.
- **Code-Related Requests**: Request code generation, modification, or explanations.
- **Open-Ended Queries**: Explore coding concepts or seek general guidance.

Contextual Prompts

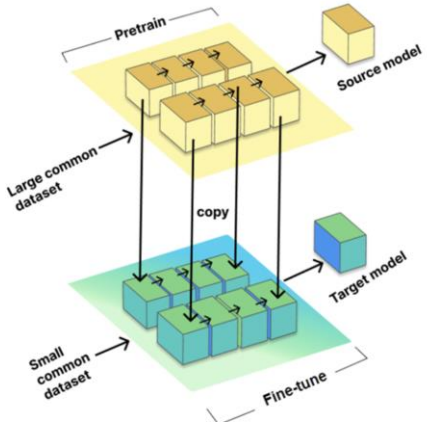
- **Code Snippets**: Provide snippets for tailored assistance.
- **Scenario Descriptions**: Describe coding scenarios for specific help.

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GitHub Copilot Large Language Models (LLMs)

Fine-tuning LLMs involves training the model on a smaller, task-specific dataset, known as the target dataset, while using the knowledge and parameters gained from the source model.

- **Volume of Training Data**: LLMs are trained on vast amounts of text from diverse sources, providing a broad understanding of language and context.
- **Contextual Understanding**: They generate contextually relevant and coherent text, making meaningful contributions to various forms of communication.
- **Machine Learning Integration**: LLMs are neural networks with millions or billions of parameters, fine-tuned to understand and predict text effectively.
- **Versatility**: These models can be tailored for specialized tasks, making them highly versatile across different domains and languages.



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Module 4: Using advanced GitHub Copilot features



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Advanced GitHub Copilot features

When GitHub Copilot is enabled, it provides you with suggestions. These suggestions are called ghost text. You can either ignore the ghost text or accept it by pressing the **Tab** key.

- **Ghost Text Suggestions**: GitHub Copilot provides real-time code suggestions called ghost text, which you can accept with the Tab key or ignore.
- **Contextual Prompts**: By default, Copilot uses open files as context but you can also provide prompts via comments, chat window, or inline chat.
- **Interactive Chat**: Use the chat feature in Visual Studio Code to ask questions about your code or other software-related queries.
- **Inline Chat**: Access inline chat with **Ctrl+i** (Windows) or **Command+i** (Mac) to interact with Copilot without leaving your code.
- **Slash Commands**: Utilize slash commands in the chat pane or inline chat for quick solutions to common development tasks.
- **Agents**: Use agents like @terminal and @workspace in Visual Studio Code to ask context-specific questions about your terminal or entire project.

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Applied GitHub Copilot techniques

- **Implicit Prompts**
  - GitHub Copilot includes features that implicitly provide a precrafted prompt to get a good answer.
  - Slash commands can be used to both in the inline chat and the chat interface.
- **Selective Context**
  - Workspace Suggestions: Customize Copilot to provide suggestions based on the entire workspace or terminal output.
  - Contextual Assistance: Use agents like @workspace to get tailored suggestions for tasks like creating Dockerfiles.
  - Use @docker to learn about containerization, for generating Docker assets for local development workflows, or analyzing project vulnerabilities
- **Combining Features**
  - Slash Commands and Inline Chat: Combine these features to interact with Copilot in the most effective way for your coding needs.
  - Flexible Interaction: Choose the method that works best for you and your code, whether it's inline chat or dedicated chat panes.
- **Rewording Prompts**
  - Refining Requests: If initial suggestions aren't satisfactory, reword your prompts to be more specific.
  - Iterative Improvement: Continuously refine your prompts or start writing code for Copilot to autocomplete, ensuring better results.

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Module 5: GitHub Copilot Across Environments: IDE, Chat, and Command Line Techniques



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Code completion with GitHub Copilot

Copilot offers code suggestions as you type: sometimes completing the current line, sometimes suggesting a whole new block of code.

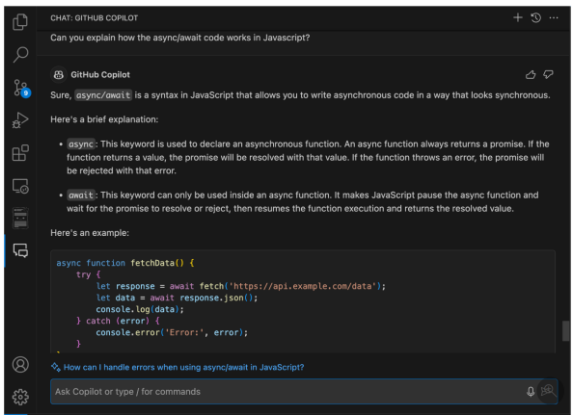
- **Supported Languages**: GitHub Copilot excels in Python, JavaScript, TypeScript, Ruby, Go, C#, and C++, but also supports many other languages and frameworks.
- **Auto Suggestions**: Provides real-time, context-aware code suggestions, completing lines or blocks of code to save development time.
- **Multiple Suggestions Pane**: Offers multiple code suggestions, allowing you to explore different options using the control panel or keyboard shortcuts.
- **Adaptability**: Adapts to your coding style, including method implementation, naming conventions, formatting, comment style, and design patterns.
- **Using Comments**: Enhances code suggestions by understanding and incorporating coding comments through natural language processing and contextual analysis.
- **Efficiency**: Helps streamline coding tasks by reducing the need to search for syntax or troubleshoot logic.

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GitHub Copilot Chat

GitHub Copilot Chat allows developers to have natural language conversations about their code, ask questions, and receive intelligent responses and suggestions in real-time.

- **Interactive AI Assistant**: GitHub Copilot Chat allows natural language conversations within the development environment for real-time coding assistance.
- **Code Generation and Debugging**: Helps generate complex code, debug errors, and provide step-by-step explanations.
- **Code Explanations**: Breaks down complex code snippets, explains functionality, and offers optimization insights.
- **Slash Commands and Agents**: Use slash commands and custom agents to perform specific actions and improve prompt effectiveness.



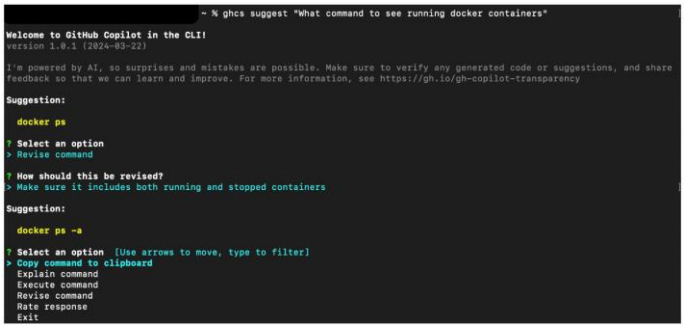
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GitHub Copilot for the Command Line

GitHub Copilot Chat allows developers to have natural language conversations about their code, ask questions, and receive intelligent responses and suggestions in real-time.

- **Command-Line Integration**: GitHub Copilot enhances command-line workflows by providing explanations, suggesting commands, and executing them.
- Offers frequently used commands like explaining "sudo apt-get" or suggesting commands to undo the last commit.
- Simplifies complex tasks and improves productivity for both new and seasoned command-line users.



See: gh copilot suggest

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Module 6: Management and customization considerations with GitHub Copilot

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Explore GitHub Copilot plans and their associated management and customization features

Management Policies: GitHub Copilot offers different features across Free, Pro, Business, and Enterprise plans, including public code filters and user management.

Security and Privacy: Business and Enterprise plans provide robust privacy controls, such as excluding specific files from analysis, accessing detailed audit logs.

Plan Selection Considerations: Key factors include data privacy, policy management, data collection and retention, and IP indemnity, which are crucial for compliance and security.

Feature	Free* & Pro	Business	Enterprise
Public code filter	✓	✓	✓
User management	✗	✓	✓
Data excluded from training by default	✓	✓	✓
Enterprise-grade security	✗	✓	✓
IP indemnity	✗	✓	✓
Content exclusions	✗	✓	✓
SAML SSO authentication	✗	✓	✓
Require GitHub Enterprise Cloud	✗	✗	✓
Usage metrics	✗	✓	✓

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Explore contractual protections in GitHub Copilot and disabling matching public code

To help ensure that your organization remains compliant with legal requirements, GitHub Copilot offers:

IP indemnity: The GitHub Copilot Business and Enterprise plans include IP indemnity, which provides legal protection against intellectual property claims related to the use of Copilot suggestions.

- With IP indemnity, if any suggestion from GitHub Copilot is challenged as infringing on third-party IP rights, GitHub assumes legal responsibility.

Data Protection Agreement (DPA): GitHub offers a DPA that outlines the measures taken to protect your data and ensure compliance with data privacy regulations.

- These agreements provide transparency and assurance that your data is handled securely and responsibly.

GitHub Copilot Trust Center: The GitHub Copilot Trust Center provides detailed information about how GitHub Copilot works, including security, privacy, compliance, and intellectual property safeguards.

- This resource helps organizations feel confident using GitHub Copilot while adhering to best practices and legal requirements.

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Manage content exclusions

The content exclusion feature in GitHub Copilot helps protect sensitive information by preventing the use of specific files, directories, or repositories to inform code-completion suggestions.

Configuring Content Exclusions for Repositories

- **Repository Settings**: Go to the repository's main page, select Settings, and specify files or directories to exclude under the Copilot section.
- **Exclusion Paths**: Use the Repositories and paths to exclude section to define exclusions.

Configuring Content Exclusions for Organizations

- **Organization Settings**: Access your organization's settings from your profile photo, then navigate to Copilot > Content exclusion.
- **Exclusion Details**: Enter the details of files or repositories to exclude from Copilot suggestions.

Impact of Content Exclusions

- **Code Completion**: Excluded files won't have code completion available, and their content won't inform suggestions in other files.
- **Chat Responses**: Excluded content won't be used in GitHub Copilot Chat responses, affecting the quality and relevance of suggestions.

Limitations of Content Exclusions

- **IDE Limitations**: Exclusions might not apply in some IDE features, like Copilot Chat in Visual Studio Code.

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Troubleshoot common problems with GitHub Copilot

The absence of code suggestions when using Copilot can be solved by trying these troubleshooting actions:

Missing Code Suggestions

- **Internet and Updates**: Ensure a stable internet connection and update the GitHub Copilot extension.
- **IDE Compatibility**: Verify that your IDE is compatible and properly configured for GitHub Copilot.

Addressing Content Exclusion Issues

- **Delayed Application**: Reload settings in your IDE to apply content exclusions immediately.
- **Scope and Limitations**: Ensure all team members have the correct settings and be aware of IDE-specific limitations.

Improving Unsatisfactory Code Suggestions

- **Clear Context**: Provide descriptive comments and meaningful variable names.
- **Prompt Techniques**: Use specific commands and experiment with prompt lengths for better suggestions.

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Module 7: Developer use cases for AI with GitHub Copilot



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Boost developer productivity with AI

GitHub Copilot helps bridge the gap between learning and actual implementation with the following:

Accelerate Learning New Languages and Frameworks

- **Code Suggestions**: Provides context-aware snippets to illustrate the usage of unfamiliar functions and libraries.
- **Language Support**: Supports a wide range of languages, aiding smooth transitions.

Minimizing Context Switching

- **In-Editor Assistance**: Offers code suggestions directly in the IDE, reducing the need to search for solutions online.

Enhanced Documentation Writing

- **Inline Comments**: Generates relevant comments explaining complex code sections.
- **Function Descriptions**: Suggests descriptions for functions, including parameter explanations and return value details.

Improving Code Consistency

- **README Generation**: creates README files with structured content based on the codebase.
- **Documentation Consistency**: Helps maintain a consistent documentation style across the project.

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Align with developer preferences

Code Generation and Completion

- **Multiple Suggestions**: Provides various code suggestions for ambiguous scenarios, allowing developers to choose the best option.
- **Language-Specific Idioms**: Suggests idiomatic code and best practices for different programming languages.

Writing Unit Tests and Documentation

- **Test Case Generation**: Suggests relevant test cases based on function signatures and behavior, including edge cases.
- **Documentation Stubs**: Generates initial documentation stubs for functions, classes, and modules, which can be refined by developers.

Code Refactoring

- **Pattern Recognition**: Identifies common patterns and suggests more efficient or cleaner alternatives.
- **Modern Syntax Suggestions**: Recommends modern language features for evolving languages like JavaScript ECMAScript.

Debugging Assistance

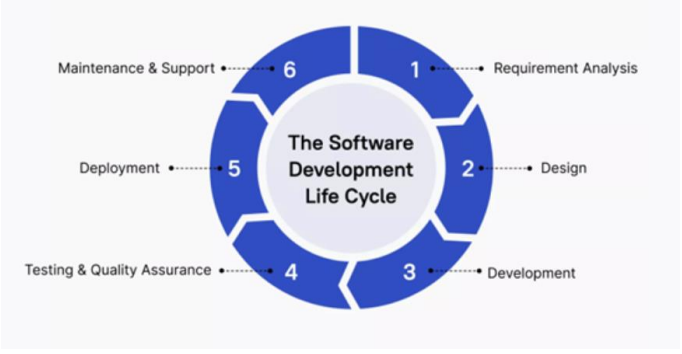
- **Error Explanation**: Provides plain-language explanations for error messages and suggests potential fixes.
- **Log Statement Generation**: Suggests relevant log statements to help diagnose issues in complex code paths.

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AI in the Software Development Lifecycle (SDLC)

GitHub Copilot enhances different SDLC phases, from initial planning to deployment and maintenance.

- **Boilerplate Code and Design Patterns**: Generates repetitive code structures and suggests appropriate design patterns.
- **Code Optimization and Translation**: Offers efficient code alternatives and assists in translating code between languages.
- **Unit Tests and Edge Cases**: Generates test cases and suggests scenarios covering edge cases.
- **Bug Fixes and Refactoring**: Proposes fixes for issues and suggests improvements to keep the codebase modern and efficient.



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Understand limitations and measure impact

Code Quality & Correctness

- **Potential for Errors**: Copilot may suggest code with bugs or that doesn't fully meet requirements.
- **Security Concerns**: Generated code might not adhere to best security practices, requiring careful review.

Language & Framework Specificity

- **Varying Performance**: Effectiveness can vary across different programming languages and frameworks.
- **Niche Technologies**: Suggestions may be less accurate or relevant for less common or newer technologies.

Complex Problem Solving

- **High-Level Design Limitations**: Copilot excels at code-level tasks but may not grasp complex architectural decisions.
- **Creativity Constraints**: While helpful, Copilot cannot replace human creativity in solving novel problems.

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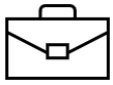
Module 8: Develop unit tests using GitHub Copilot tools



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Examine the unit testing tools and environment





GitHub Copilot Tools in Visual Studio Code

- **Code Line Completions**: Write code more efficiently with real-time suggestions.
- **Inline Chat**: Start conversations directly from the editor for coding help.
- **Smart Actions**: Run tasks without writing prompts, enhancing productivity.

Visual Studio Code Support for Unit Tests

- **Required Resources**: .NET 8.0 SDK or later, C# Dev Kit extension, and a test framework package.
- **Test Explorer**: View all test cases in a tree view.
- **Run/Debug Tests**: Features to run and debug test cases, and view results.

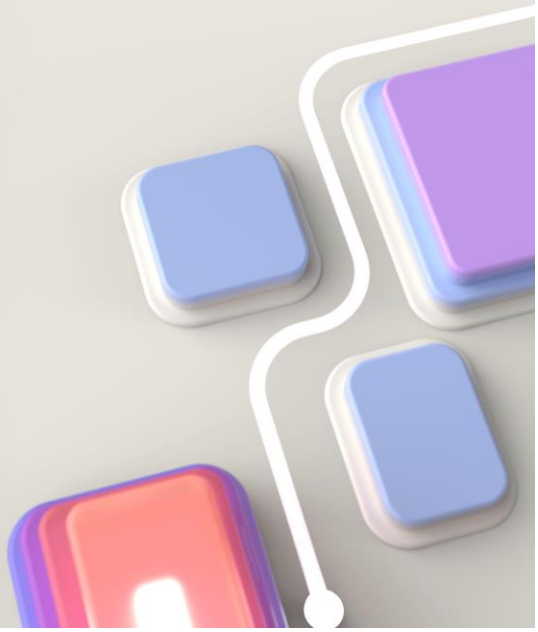
Enabling a Test Framework

- **Command Palette**: Use Ctrl+Shift+P (Windows/Linux) or Cmd+Shift+P (macOS) to open the Command Palette.
- **Project Setup**: Create new test projects and configure test runners using the Command Palette.
- **Framework Support**: Supports xUnit, NUnit, and MSTest for creating and managing unit tests.

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Module 9: Introduction to GitHub Copilot Business



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About GitHub Copilot for Business

GitHub Copilot Business builds the power of generative AI into an editor extension that suggests code in real time.

Features in Copilot Business include:

- **A more powerful AI model:** New modeling algorithms improve the quality of code suggestions.
- **AI-based security vulnerability filtering:** GitHub Copilot Business automatically blocks common insecure code suggestions by targeting issues such as hard-coded credentials, SQL injections, and path injections.
- **VPN proxy support:** GitHub Copilot works with VPNs including self-signed certificates, allowing developers to use it in any working environment.
- **Simple sign-up:** Any company can quickly purchase Copilot Business licenses online and easily assign users, even if they don't use the GitHub platform for their source code.

Features	GitHub Copilot Business	GitHub Copilot Pro & Free*
Exclude specific files from GitHub Copilot	✓	✗
Organization-wide policy management	✓	✗
Audit logs	✓	✗
Increase GitHub Models rate limites	✓	✗

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GitHub Copilot for Business use cases and customer stories

Faster Coding

- **Increased Speed**: Nearly 90% of developers complete tasks faster with GitHub Copilot.
- **Time Savings**: Coyote Logistics saved significant time on tasks like writing Terraform configurations.
- **Efficient Details**: GitHub Copilot fills in details after the base configuration is written.

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Improved Flow

- **Reduced Disruptions**: 73% of developers said Copilot helps them stay in the flow longer.
- **Mental Energy**: 87% of developers preserved mental energy when working through repetitive tasks.
- **Focus Maintenance**: Helps developers stay focused on solving complex business challenges.

Enhanced Productivity

- **Context Switching**: Duolingo reduced context switching and manual boilerplate code production.
- **New Developers**: Estimated 25% speed increase for developers new to a repository or framework.

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How to get started with GitHub Copilot for Business

Enforcing a Policy

- **Access Policies**: In the enterprise sidebar, select Policies, then Copilot, and configure access for your subscription.
- **Manage Access**: Under Manage organization access to GitHub Copilot, set the desired access level.

Enabling Access for All Users

- **Organization Settings**: Select your profile photo, then Your organizations, and navigate to Settings.
- **User Permissions**: In the Copilot Access section, select Allow for all members, confirm seat assignment, and save changes.

Enabling Access for Selected Users

- **Selective Access**: Follow the enabling steps, then select Selected teams/users under User permissions.
- **Confirm and Save**: Confirm seat assignment for selected users and save changes.

Disabling Access for the Organization

- **Disable Access**: Follow the enabling steps, then select Disabled under User permissions.
- **Confirm and Save**: Confirm the decision to disable access for all users and save changes.

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Module 10: Introduction to GitHub Copilot Enterprise



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About GitHub Copilot Enterprise

- GitHub Copilot Enterprise includes collaborative chat within pull requests, pull request summaries, docset management, and code review.
- Enterprise and Business plans offer zero data retention for code snippets, organization-wide policy management, and VPN proxy support, with Enterprise providing comprehensive security tool integration.
- All plans integrate with your editor and live on GitHub.com, but only Enterprise and Business plans offer advanced management features.

	GitHub Copilot Enterprise	GitHub Copilot for Business	GitHub Copilot Pro & Free
Collaborative Chat within pull requests	✓	×	×
Lives on Github.com	✓	✓	✓
Plugs right into your editor	✓	✓	✓
Pull request summaries	✓	✓	✓
Copilot docset management	✓	×	×
Copilot code review	✓	×	✓
Zero data retention for code snippets and usage telemetry	✓	✓	×
Organization-wide policy management	✓	✓	×
Integration with security tools	✓	Limited	×
Audit logging and reporting	✓	Limited	×
VPN proxy support via self-signed certificates	✓	✓	×

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How to get started

Your enterprise owner can enable GitHub Copilot Enterprise for the organizations in the enterprise by first establishing the policy and then assigning users.

Enforce a Policy

- **Access Policies**: In the enterprise sidebar, select Policies, then Copilot, and configure access for your subscription.
- **Manage Access**: Under Manage organization access to GitHub Copilot, set the desired access level.

Enable Access for All Users

- **Organization Settings**: Select your profile photo, then Your organizations, and navigate to Settings.
- **User Permissions**: In the Copilot Access section, select Allow for all members, confirm seat assignment, and save changes.

Enable Access for Selected Users

- **Selective Access**: Follow the enabling steps, then select Selected teams/users under User permissions.
- **Confirm and Save**: Confirm seat assignment for selected users and save changes.

Disable Access for the Organization

- **Disable Access**: Follow the enabling steps, then select Disabled under User permissions.
- **Confirm and Save**: Confirm the decision to disable access for all users and save changes.

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Module 11: Using GitHub Copilot with JavaScript



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What is GitHub Copilot

To use GitHub Copilot, you need to complete the following steps:

1. **GitHub Account**:
   - Create a GitHub account. Since Copilot is a GitHub service, you need a GitHub account to use it.
2. **Sign Up for Copilot**:
   - Sign up for Copilot via its web page.
3. **Enable Copilot**:
   - In GitHub, go to your profile, then head over to **Settings**.
   - Under **Copilot**, enable access or sign up for GitHub Copilot Free or try a trial for one of the tiered plans.
4. **Install the Extension**:
   - Install the GitHub Copilot extension into your IDE.
   - Extensions are available for major IDEs like Visual Studio and Visual Studio Code.

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Use GitHub Copilot with JavaScript

The quality of output from Copilot is dependent on how well you craft your prompt.

Prompt Engineering

- Build specific and clear prompts to generate useful suggestions from GitHub Copilot.
- Provide context and examples to ensure high-quality output from Copilot.

Best Practices

- **Simple to Complex**: Start with simple prompts and gradually add more details for specific suggestions.
- **Cycle Suggestions**: Use **Ctrl+Enter** (**Cmd+Enter** on Mac) to cycle through various suggestions and choose the best one.

Using GitHub Copilot Chat

- **Interactive Interface**: Use the chat interface to ask and receive answers to code-related questions.
- **Prompt Input**: Add prompts in the chat input to interact with the output and refine suggestions.

Continuous Updates

- **Code Freshness**: Ensure your code is updated and fresh by fixing bugs and adding new features.
- **Usability Improvements**: Use Copilot to enhance functionality and usability in your projects.

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Module 12: Using GitHub Copilot with Python



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What is GitHub Copilot?





How GitHub Copilot Works

- **AI Assistant:** GitHub Copilot generates code and more based on natural language prompts typed within your IDE.
- **Prompts:** A natural language prompt that can generate code for you to use looks like:

```
# Create a web API using FastAPI
with a route to products
```

Recognizing Prompts

- **Comment in Code File:** Copilot recognizes prompts typed as comments in code files (e.g., .py, .js).
- **Markdown File:** Typing text in a markdown file and waiting a few seconds will also prompt Copilot to respond (configurable).

Accepting Suggestions

- **Gray Code:** *Suggestions* appear as gray code; press the **Tab** key to accept.
- **Multiple Suggestions:** Cycle through and select the most appropriate suggestion (macOS: Atl +], Windows and Linux: Atl +])

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Use GitHub Copilot with Python





Developing with GitHub Copilot

- **Continuous Updates:** Ensure code is fresh, fix bugs, and add new features using GitHub Copilot and Copilot Chat.
- **Interactive Chat:** Use Copilot Chat to ask and receive answers to code-related questions.

Prompt Engineering

- **Creating Prompts:** Build prompts with clear instructions to generate specific responses from Copilot.
- **Effective Prompts:** Crafting specific and clear prompts ensures high-quality output from Copilot.

Best Practices for Using Copilot

- **Simple to Elaborate:** Start with simple prompts and gradually add more details for specific suggestions.
- **Cycle Suggestions:** Use **Ctrl+Enter** to cycle through suggestions and select the best output.

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